

Estimate and Predict the Foreign Currency Exchanging Rates by Using Adaptive Neuro Fuzzy Inference System and Deep Learning Algorithms

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Abstract—To correctly and accurately predict and estimate the exchanging rates among various currencies to get the maximum profit is a challenging task, and it is critical important to all financial institutions under the current fluctuation situation. In this study, we try to use different AI methods and algorithms, such as Adaptive Neuro Fuzzy Inference System (ANFIS) and Deep Learning (DL), to easily and correctly predict and estimate the current currency exchanging rate. Combining with some appropriate pre-data-processing techniques, the current currency exchanging rates could be accurately and quickly estimated via those models. In this research, both algorithms are designed and built to help decision makers working in the financial institutions to easily and conveniently predict the current exchanging rates. The minimum training and checking RMSE values for ANFIS model can be 0.0009828 and 0.001713. The minimum MSE value for DL model is 0.0000047 with a regression value of 0.9958.

Keywords- ANFIS algorithm; deep learning; estimate and predict exchanging rates for currencies; AI applications in financial implementations.

I. INTRODUCTION

As the fast development of AI technologies, such as fuzzy inference systems, machine learning and deep learning, today various AI related algorithms have been widely implemented in financial fields to estimate and predict the stock values, currency exchanging rates, bonus analyses and all other related applications [1-7].

Most of researches are concentrated on stock predictions or estimations based on neural networks, machine learning, and deep learning studies. Different and various machine learning algorithms accompanied with some sophisticated additions are applied on stock analyses and predications to improve the accuracy of prediction on stock markets. L. S. Chong *et al.* reported to use Ensemble of Deep Neural Networks to performance prediction for stock markets [8]. L. Yu developed an algorithm based on deep learning and neural networks to improve the analyses for economic and financial data [9]. V. Polepally *et al.* and Y. K. Pardeshi and P. P. Kale reported to use machine learning and deep learning algorithms to improve the prediction accuracy for current stock prices [10-11]. H. J. Singh, *et al.* and Y. Lin *et al.* developed a novel multivariate recurrent neural network and a new convolutional neural network with long short term

memory combined model to estimate the current stock prices and their tendency [12-13]. K. Singh *et al.* performed a comparative studies and analysis for different stock price prediction techniques developed in recent years [14]. S. Roy and S. Tanveer developed an algorithm to forecast stock price by using DeepNet method [15]. Instead of using any traditional machine learning model, M. Tarsi *et al.* utilized a Long Short Term Memory (LSTM), which is a variation of machine learning model, to predict the stock price [16]. Mandee, A *et al.* utilized an explainable artificial intelligence XAI to predict stock market trends [17].

K. Chinprasatsak *et al.* reported to use neural network for forecasting high and low price on foreign exchange market [18]. A. Alamsyah and W. H. Aprillia and P. Aggarwal and A. K. Sahani performed some studies in comparisons of the foreign currency prediction performance with neural network algorithms [19-20]. T. A. Bui *et al.* and G. Singh *et al.* reported to use neural networks and CNN-RNN based hybrid machine learning model to predict the currency exchange rate [21-22]. E. Sarma *et al.* performed a comparison study among different machine learning classification methods used for currency exchange rate trends [23]. A. L. C. Tak and R. Logeswaran also developed a foreign currency prediction method based on machine learning technique [24].

Comparably speaking, there are less studies related to the currency exchanging rate based on neural network or machine learning have been reported in recent years compared with those studies on the stock price predictions. Also almost all those studies are only related to using single method or algorithm to perform predictions. In this study, we will concentrate on the prediction and estimation of the foreign currency exchanging rate based on two different AI related techniques; Adaptive Neuro Fuzzy Inference System (ANFIS) and Deep Learning (DL) algorithm.

To correctly and accurately predict and estimate the currency exchanging rates to get the maximum profit via different AI methods, some correct AI models are necessary with popular algorithms, such as Adaptive Neuro Fuzzy Inference System (ANFIS) and Deep Learning (DL). Combining with some appropriate pre-data-processing techniques, the current currency exchanging rates could be accurately and quickly estimated via those models. In this research, both algorithms are designed and built to help

decision makers working in the financial institutions to easily and conveniently predict the current exchanging rates.

Currency exchanging rates are changed at any moment, and they may be varied significantly day by day, month by month and year by year. Due to the heavy complicity and unforeseen variations on the current market, to correctly and accurately predict the currency exchanging rate needs the following factors and operational steps to be taken:

- 1) The changing or variation of the currency exchanging rates can be considered as a periodic function, and this period could be 3 months, 6 month or longer, which depends on the target period on each research. In our case, we used 3 month as a period.
- 2) Based on assumption above, we utilized the NASDAQ dataset to train and check our target ANFIS and DL models.

In this study, we only take care of the prediction for one of the most popular currency exchanging rate transactions, Euro to US Dollar conversions. This study is divided into 6 sections; after this Introduction, an introduction to two NASDAQ datasets used as the training and checking for AI models is provided in section 2. The ANFIS and its implementations is discussed in section 3. A discussion about DL is given in section 4. The experiment studies and results are given in section 5. The conclusion and future works are provide in section 6

II. INTRODUCTION TO NASDAQ CURRENCY EXCHANGING RECORDS DATASET AND DATA PREPROCESSING

Two NASDAQ datasets [25], one contained 6-month transactions including currency exchanging rates from May 6, 2022 to Nov. 4 2022 and the other included 3-month transaction records from Feb. 6 2023 to May 5 2023, are utilized in this study. Both of them are about Euro to US Dollar transactions. The first one is used as the training and checking data for ANFIS and DL models, and the second works as the testing and validation purpose for those models.

Each dataset contained six columns, Date, Open, High, Low, Volume and Close, with either 6-month or 3-month currency exchanging records. Each column can be mapped to the Opening rate, Highest rate, Lowest rate, transaction Volume and Closing rate. For our study, we only need four of them; Open, High, Low and Close. In fact, we use the first three columns, Open, High, Low as inputs and the Close column as the output.

A crystal key issue in using those data to train, check and test our ANFIS or DL models is the data preprocessing. As everybody know, the exchanging rates are changed or varied in every moment, not each day, and the mounts they changed are significant in a relatively wide range, or even dynamically, for a period of time. This provided a challenging issue when using ANFIS, especially using the fuzzy rules, to estimate the output or the closing rate due to the significant variations in the rate values. In the worst case,

the ANFIS could not perform its FIS function due to the out-of-bound of the input values with too big different rate values for different time period.

To effectively solve that key issue, we need to preprocess those data, exactly to perform a normalization job for those data to enable them to be used in our model training and checking. In summary, we only take care of those relative changing values on the rates, but not for the absolute changing rates, which is good enough for us since we only pay our attention to the changing rate values in trends or tendency.

III. ADAPTIVE NEURO FUZZY INFERENCE SYSTEM (ANFIS)

The so-called ANFIS is exactly a combination of two soft-computing techniques: Artificial Neural Network (ANN) and Fuzzy Inference System (FIS), which was first introduced by Jyh-Shing Roger Jang in 1992 [26]. The FIS used a Sugeno fuzzy inference system and its structure is similar to a multilayer feed forward neural network structure, but the difference is that the links between nodes in ANFIS define the signals' flow direction and there are no associated weight factors with the links. It consists of a network of neurons that communicate between the input and hidden layers, as well as the hidden and output layers.

Each layer consists of neurons constructed according to the principles of fuzzy control. Figure 1 shows a Sugeno fuzzy model with 27 rules along with a corresponding ANFIS architecture. In our case, total 27 rules in the method of "If-Then" for the Sugeno model are considered with x and y as inputs and f as output [27]. 27 rules are defined as below (three input columns – *Open*, *High*, *Low*, L: value low, M: value mid, H: value high):

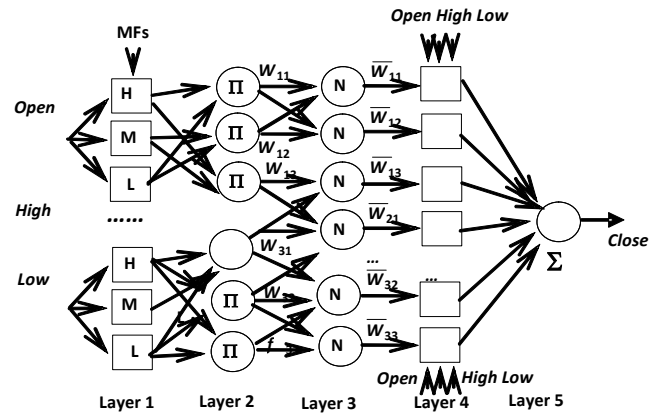


Figure 1. A typical structure of the ANFIS.

- R_1 : If *Open* is L and *High* is L, and *Low* is L,
then $f_{111} = p_{111}Open + q_{111}High + r_{111}Low + c_{111}$
- R_2 : If *Open* is L and *High* is L, and *Low* is M,
then $f_{112} = p_{112}Open + q_{112}High + r_{112}Low + c_{112}$
- R_3 : If *Open* is L and *High* is M, and *Low* is L,
then $f_{21} = p_{113}Open + q_{113}High + r_{113}Low + c_{113}$

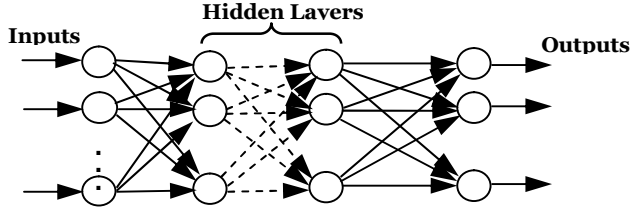


Figure 2. A model of multilayer feed-forward ANN or DL.

R₄: If *Open* is L and *High* is M, and *Low* is M,
then $f_{22} = p_{211} \text{Open} + q_{211} \text{High} + r_{211} \text{Low} + c_{211}$
.....

IV. THE ARTIFICIAL NEURAL NETWORKS (ANN) AND DEEP LEARNING (DL)

Regularly a complete artificial neural network (ANN) has multiple nodes with multiple layers, including input layer, output layer and hidden layers. Figure 2 shows a model of multiple layers feed forward ANN or DL. The dash lines means that multilayer are included in this ANN and these layers cannot be observable. Exactly in Figure 2, on each feed forward arrow branch from one node to another, a weight factor w_{ij} should be multiplied to obtain a complete transfer signal.

Generally, a neural network can be adjusted or trained, so that a particular input leads to a corresponding target output. The network is adjusted, based on a comparison between the output and the target, until the network output matches the target. Typically, many such input-target pairs are needed to train a network, which is called a supervised learning model.

As we did for the ANFIS, the NASDAQ dataset is utilized to train, check and test this DL model. Three columns, *Open*, *High* and *Low*, work as inputs and the *Close* as the output. Totally 77 set of data are used with 70% as training data, 15% as checking data and 15% as testing data. The Levenberg-Marquard training algorithm and MATLAB Deeping Learning Toolbox are used to perform these tasks to generate our desired ANN/DL model with 15 hidden layers. The ANN/DL structure is shown in Figure 3.

V. EXPERIMENTAL STUDIES AND RESULTS

By using the NASDAQ 6-month dataset as the training and checking data to train and check our ANFIS model, the model structure and the training result is shown in Figures 4a and 4b.

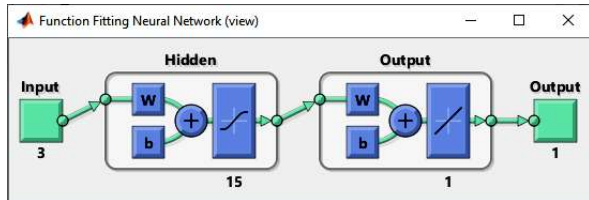
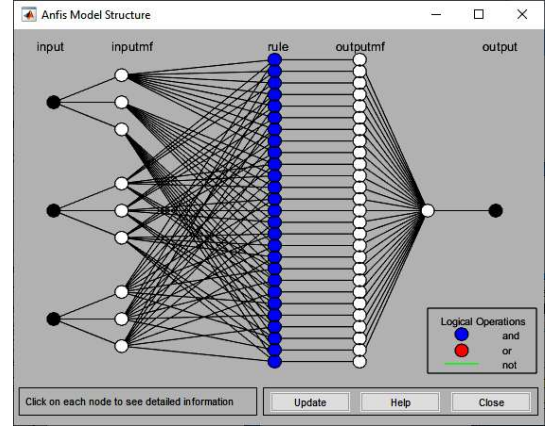
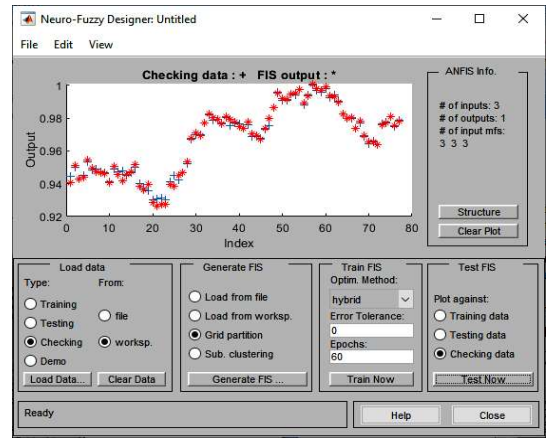


Figure 3. The structure used in our ANN/DL model.



(a)



(b)

Figure 4. The ANFIS model structure and training results

The testing and performance result for the ANN/DL model are shown in Figure 5.

A comparison study is performed for the validation errors between the ANFIS and ANN/DL model, and this comparison result is shown in Figure 6. Both validation errors are RMS values by comparing the actual output and the checking data inputs for ANFIS and ANN/DL systems used in this study.

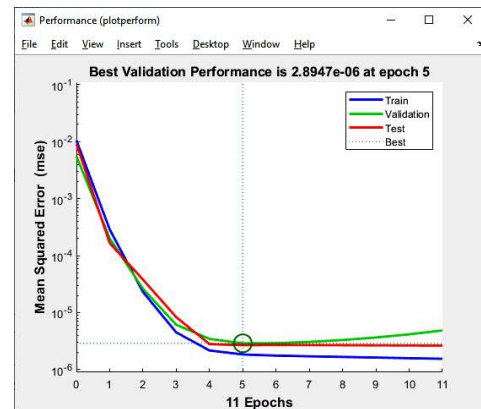
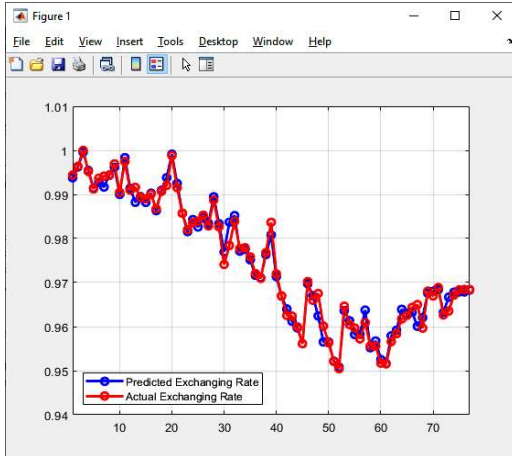
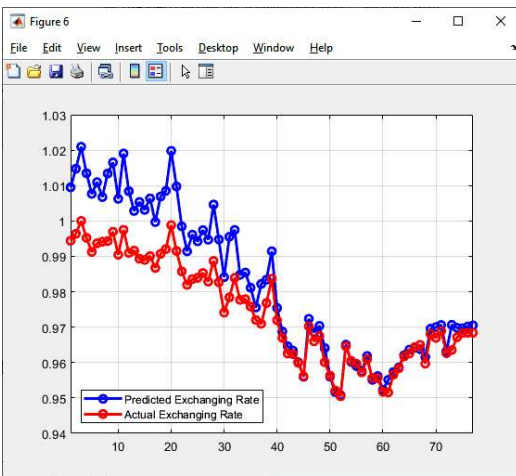


Figure 5. The validation and performance results of ANN/DL model.



(a)



(b)

Figure 6. Comparison between ANFIS and ANN/DL algorithms.

It can be found that the RMSE value for the ANFIS method is about 0.0016, and the RMSE value for ANN/DL model is 0.0044 with a regression value of 0.9954. The ANFIS algorithm result is better compared with the ANN/DL algorithm for our study, exactly the ANFIS algorithm is about 64% better than that of ANN/DL algorithm by checking the validation RMSE error values in Figure 6 for our study.

VI. CONCLUSION AND FUTURE WORKS

With the help of MATLAB Fuzzy Logic and Deep Learning Toolboxes as well as NASDAQ dataset, we develop two AI models with two related algorithms, ANFIS and ANN/DL, to perform prediction for foreign currency exchanging rates. First we utilized NASDAQ 6-month dataset to train ANFIS and ANN models, the first 3-month data is used to training, and the second 3-month data is used to checking models. To confirm and check the effectiveness and accuracy, we utilized another NASDAQ 3-month dataset to validate both models. A comparison result shows that the ANFIS model overtakes the ANN/DL model for our study.

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